Chapter 2. ARMA Models

Autoregressive and moving average processes, PACF

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2.1 Introduction

Motivation. ARMA models build stationary dependence from i.i.d. noise by summing neighbouring noise terms (MA) and regressing on the past (AR).

Let $\epsilon_1, \epsilon_2, \dots$ be i.i.d. with mean 0 and variance σ^2.

2.1.1 Summing noise: MA(1)

For a real θ put

$$X_t = \epsilon_t + \theta \epsilon_{t-1}. \tag{2.1}$$

Then $EX_t = 0$ and, since $\mathrm{Cov}(\epsilon_s, \epsilon_t) = \sigma^2 \mathbb{1}\{s = t\}$,

$$\gamma(h) = \mathrm{Cov}(\epsilon_t + \theta\epsilon_{t-1},\, \epsilon_{t+h} + \theta\epsilon_{t+h-1}) = \begin{cases} (1+\theta^2)\sigma^2, & h = 0, \\ \theta\sigma^2, & |h| = 1, \\ 0, & |h| \geq 2, \end{cases}$$

free of t: the stationary *moving average model of order* 1, MA(1).

2.1.2 Regressing on the past: AR(1)

Consider

$$X_t = \phi X_{t-1} + \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2). \tag{2.2}$$

Case $\phi = 1$ (random walk). $X_t = X_0 + \epsilon_1 + \cdots + \epsilon_t$, so $\mathrm{Var}(X_t - X_0) = t\sigma^2$. A stationary solution would give

$$\mathrm{Var}(X_t - X_0) = 2\gamma(0) - 2\gamma(t), \tag{2.3}$$

bounded by $4\gamma(0)$ (Theorem 1.2.6(3)), while $t\sigma^2 \to \infty$. Hence no stationary solution exists. For $\phi = -1$, $X_{2t} - X_0 = \sum_{j=1}^{2t} (-1)^j \epsilon_j$ has variance $2t\sigma^2$: same contradiction.

Case $|\phi| < 1$. If a stationary solution exists, iterating (2.2),

$$X_t = \phi X_{t-1} + \epsilon_t = \phi(\phi X_{t-2} + \epsilon_{t-1}) + \epsilon_t = \cdots = \sum_{j=0}^{n-1} \phi^j \epsilon_{t-j} + \phi^n X_{t-n}. \tag{2.4}$$

As $E(\phi^n X_{t-n})^2 = \phi^{2n} EX_0^2 \to 0$, the only candidate is

$$X_t = \sum_{i=0}^{\infty} \phi^i \epsilon_{t-i} \tag{2.5}$$

(convergent in L^2 and a.s.). It is a solution,

$$\phi X_{t-1} + \epsilon_t = \sum_{i=0}^{\infty} \phi^{i+1} \epsilon_{t-1-i} + \epsilon_t = \sum_{j=1}^{\infty} \phi^j \epsilon_{t-j} + \epsilon_t = X_t,$$

and stationary: $EX_t = 0$ and

$$\mathrm{Cov}(X_t, X_{t+h}) = \mathrm{Cov}\Big(\sum_{j=0}^{\infty} \phi^j \epsilon_{t-j}, \sum_{k=0}^{\infty} \phi^k \epsilon_{t+h-k}\Big) = \sum_{j=0}^{\infty} \phi^j \phi^{j+h} \sigma^2 = \phi^h \frac{\sigma^2}{1-\phi^2}, \quad h \geq 0.$$

This is the *AR(1) model*.

2.2 Autoregressive models

Motivation. AR(p) regresses X_t on p lags; stationarity is decided by the roots of a polynomial.

Definition 2.2.1 (Autoregressive model)

If $\{X_t\}$ satisfies

$$X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2} - \cdots - \phi_p X_{t-p} = \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \tag{2.6}$$

then $\{X_t\}$ follows an *autoregressive model of order* p, written $\{X_t\} \sim AR(p)$.

The *characteristic polynomial* is

$$\phi(z) = 1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_p z^p. \tag{2.7}$$

If every root of ϕ satisfies $|z| > 1$, then on $|z| \le 1$

$$\frac{1}{\phi(z)} = \sum_{i=0}^{\infty} a_i z^i, \tag{2.8}$$

and the stationary solution of (2.6) is

$$X_t = \sum_{i=0}^{\infty} a_i \epsilon_{t-i}. \tag{2.9}$$

X_t depends only on ϵ_s, $s \le t$, hence is independent of ϵ_s, $s > t$: a *causal* representation (Proposition 2.2.B).

2.2.1 The backshift operator

$BX_t = X_{t-1}$, $Bf(t) = f(t-1)$, $B^j X_t = X_{t-j}$. For $a(z) = \sum_{i\ge 0} a_i z^i$, $b(z) = \sum_{j\ge 0} b_j z^j$ with $\sum |a_i|, \sum |b_j| < \infty$,

$$c(z) = (ab)(z) = a(z)b(z) = \sum_{k=0}^{\infty} c_k z^k, \qquad c_k = \sum_{i=0}^{k} a_i b_{k-i}.$$

For stationary $\{X_t\}$,

$$a(B)\big(b(B)X_t\big) = (ab)(B)X_t = b(B)\big(a(B)X_t\big), \tag{2.10}$$

all stationary. Thus (2.6) is $\phi(B)X_t = \epsilon_t$ and (2.9) is

$$X_t = \frac{1}{\phi(B)}\epsilon_t, \qquad \frac{1}{\phi(B)} = \sum_{i=0}^{\infty} a_i B^i. \tag{2.11}$$

Lemma 2.2.A (Operator calculus; supplement)

Let $\{X_t\}$ satisfy $\sup_t E|X_t| < \infty$ (e.g. stationary), and let $\sum |a_i| < \infty$, $\sum |b_j| < \infty$. Then $b(B)X_t = \sum_j b_j X_{t-j}$ converges absolutely a.s. and in L^1, (2.10) holds a.s., and if $\{X_t\}$ is stationary so is $\{b(B)X_t\}$, with ACVF $\sum_{j,k} b_j b_k \gamma_X(h + j - k)$.

Proof. [Supplement]

Step 1. With $M = \sup_t E|X_t|$, $E\sum_j |b_j||X_{t-j}| \le M \sum_j |b_j| < \infty$ (monotone convergence): absolute a.s. and L^1 convergence. If stationary, Minkowski gives $\|\sum_j |b_j||X_{t-j}|\|_2 \le \sum_j |b_j| \, \|X_0\|_2$: L^2 convergence.

Step 2. $\sum_{i,j}|a_i||b_j|E|X_{t-i-j}| \le M\sum_i|a_i|\sum_j|b_j| < \infty$, so $\sum_{i,j}a_ib_jX_{t-i-j}$ converges absolutely a.s. and can be summed in any order: over j first gives $a(B)(b(B)X_t)$, by $k=i+j$ gives $(ab)(B)X_t$; $ab=ba$ gives the third form.

Step 3. $E\,b(B)X_t = \mu\sum b_j$, and by L^2 convergence $\mathrm{Cov}(b(B)X_t, b(B)X_{t+h}) = \sum_{j,k}b_jb_k\gamma_X(h+j-k)$, free of t. ■

Proposition 2.2.B (Causal stationary solution; supplement)

Suppose $\phi(z)\neq 0$ for all $|z|\le 1$. Then:
(1) there are $r>1$ and $C<\infty$ with $1/\phi(z)=\sum_{i\ge 0}a_iz^i$ for $|z|<r$ and $|a_i|\le Cr_0^{-i}$ for any fixed $1<r_0<r$;
(2) $X_t=\sum_i a_i\epsilon_{t-i}$ is a stationary solution of (2.6);
(3) it is the only stationary solution.

Proof. [Supplement]

Step 1. Let $r>1$ be the smallest root modulus of ϕ. Then $1/\phi$ is analytic on $|z|<r$, and Cauchy's estimate on $|z|=r_0$ gives $|a_i|\le r_0^{-i}\max_{|z|=r_0}|1/\phi(z)| = Cr_0^{-i}$; so $\sum|a_i|<\infty$.

Step 2. By the Lemma, $X_t=a(B)\epsilon_t$ is stationary and $\phi(B)X_t=(\phi a)(B)\epsilon_t=\epsilon_t$, as $\phi a=1$ on $|z|\le 1$.

Step 3. If $\phi(B)Y_t=\epsilon_t$ with $\{Y_t\}$ stationary, then by the Lemma

$$a(B)\epsilon_t = a(B)(\phi(B)Y_t) = (a\phi)(B)Y_t = Y_t,$$

so $Y_t=X_t$ a.s. ■

For $p=1$, the root $1/\phi$ lies outside the unit circle iff $|\phi|<1$, and $a_i=\phi^i$ recovers (2.5).

Remark 2.2.2 (Nonzero mean)

A stationary AR(p) model with mean μ is written

$$X_t-\mu-\phi_1(X_{t-1}-\mu)-\phi_2(X_{t-2}-\mu)-\cdots-\phi_p(X_{t-p}-\mu)=\epsilon_t, \qquad \{\epsilon_t\}\sim IID(0,\sigma^2).$$

Example 2.2.3 (Variance of a stationary AR(1))

Let $X_t=\phi X_{t-1}+\epsilon_t$ with $|\phi|<1$. As $\mathrm{Cov}(X_{t-1},\epsilon_t)=0$,

$$\gamma(0)=\mathrm{Var}(\phi X_{t-1}+\epsilon_t)=\phi^2\gamma(0)+\sigma^2, \qquad \text{hence} \qquad \gamma(0)=\frac{\sigma^2}{1-\phi^2}.$$

2.3 Moving average models

Motivation. MA(q) models are always stationary, and their ACF vanishes beyond lag q.

Definition 2.3.1 (Moving average model)

If $\{X_t\}$ satisfies

$$X_t=\epsilon_t+\theta_1\epsilon_{t-1}+\cdots+\theta_q\epsilon_{t-q}, \qquad \{\epsilon_t\}\sim IID(0,\sigma^2), \tag{2.12}$$

then $\{X_t\}$ follows a *moving average model of order* q, written $\{X_t\}\sim MA(q)$.

With the backshift operator,

$$X_t = \theta(B)\epsilon_t, \qquad \theta(z) = \sum_{i=0}^{q} \theta_i z^i, \quad \theta_0 = 1. \tag{2.13}$$

For MA(1) with $\{\epsilon_t\} \sim WN(0, \sigma^2)$ (§2.1),

$$\gamma(h) = \mathrm{Cov}(\epsilon_t + \theta\epsilon_{t-1}, \epsilon_{t+h} + \theta\epsilon_{t+h-1}) = \begin{cases} (1+\theta^2)\sigma^2, & h = 0, \\ \theta\sigma^2, & |h| = 1, \\ 0, & |h| \geq 2. \end{cases} \tag{2.14}$$

In general, for $h \geq 0$,

$$\gamma(h) = \sum_{i=0}^{q}\sum_{j=0}^{q} \theta_i\theta_j \,\mathrm{Cov}(\epsilon_{t-i}, \epsilon_{t+h-j}) = \sigma^2 \sum_{i=0}^{q-h} \theta_i\theta_{i+h}$$

(empty for $h > q$), so

$$\gamma(h) = \begin{cases} 0, & |h| \geq q+1, \\ (\theta_{|h|} + \theta_1\theta_{1+|h|} + \cdots + \theta_{q-|h|}\theta_q)\sigma^2, & |h| \leq q. \end{cases} \tag{2.15}$$

An MA(q) is stationary with variance $(1 + \theta_1^2 + \cdots + \theta_q^2)\sigma^2$; a sample ACF vanishing beyond some lag suggests an MA model.

2.4 Autoregressive moving average models

Motivation. ARMA models describe rich dependence with few parameters; causality (roots of ϕ) gives stationarity and invertibility (roots of θ) recovers the noise.

Definition 2.4.1 (ARMA model)

If $\{X_t\}$ satisfies

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = \epsilon_t + \theta_1\epsilon_{t-1} + \cdots + \theta_q\epsilon_{t-q}, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \tag{2.16}$$

then $\{X_t\}$ follows an *autoregressive moving average model of orders p and q*, written $\{X_t\} \sim ARMA(p, q)$. With $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$ and $\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$, (2.16) reads

$$\phi(B)X_t = \theta(B)\epsilon_t. \tag{2.17}$$

If $\{X_t\}$ is stationary with mean $\mu \neq 0$, X_t is replaced by $X_t - \mu$ in (2.17).

Pure AR fits often need a large order; by the *principle of parsimony* (fewer parameters for equal fit) ARMA models are preferred.

If $\phi(z) \neq 0$ for all $|z| \leq 1$, $\{X_t\}$ is stationary and

$$X_t = \sum_{i=0}^{\infty} a_i\epsilon_{t-i}, \qquad \sum_{i=0}^{\infty} a_i z^i = \frac{\theta(z)}{\phi(z)}, \tag{2.18}$$

i.e. by (2.10)

$$X_t = \frac{\theta(B)}{\phi(B)}\epsilon_t = \left(\frac{\theta}{\phi}\right)(B)\epsilon_t. \tag{2.19}$$

Its ACVF decays geometrically: ARMA processes are *short-memory processes*.

Proposition 2.4.A (Causal ARMA and geometric decay; supplement)

If $\phi(z) \neq 0$ for $|z| \leq 1$, then (2.18) is the unique stationary solution of (2.17), with $|a_i| \leq C r_0^{-i}$ for some $C < \infty$, $r_0 > 1$, and

$$\gamma(h) = \sigma^2 \sum_{i=0}^{\infty} a_i a_{i+|h|}, \qquad |\gamma(h)| \leq C' r_0^{-|h|}.$$

Proof. [Supplement]

Step 1. θ/ϕ is analytic on $|z| < r$, so $|a_i| \leq C r_0^{-i}$ as in Proposition 2.2.B.

Step 2. By the Lemma, $X_t = a(B)\epsilon_t$ is stationary and $\phi(B)X_t = (\phi a)(B)\epsilon_t = \theta(B)\epsilon_t$. Any stationary solution satisfies $Y_t = (\phi^{-1}\phi)(B)Y_t = \phi^{-1}(B)\theta(B)\epsilon_t = a(B)\epsilon_t$, $\phi^{-1} = 1/\phi$ having summable coefficients.

Step 3. For $h \geq 0$, $\gamma(h) = \sum_{i,j} a_i a_j \operatorname{Cov}(\epsilon_{t-i}, \epsilon_{t+h-j}) = \sigma^2 \sum_i a_i a_{i+h}$, and

$$|\gamma(h)| \leq \sigma^2 C^2 \sum_{i \geq 0} r_0^{-i} r_0^{-i-h} = \frac{\sigma^2 C^2}{1 - r_0^{-2}} r_0^{-h}.$$

■

2.4.1 Invertibility

If $\{X_t\}$ is stationary ARMA and $\theta(z) \neq 0$ for all $|z| \leq 1$, then

$$\sum_{j=0}^{\infty} \pi_j X_{t-j} = \epsilon_t, \qquad \sum_{j=0}^{\infty} \pi_j z^j = \frac{\phi(z)}{\theta(z)}, \quad \pi_0 = 1, \tag{2.20}$$

and $\{X_t\}$ is *invertible*: the noise is a linear combination of past observations.

Proposition 2.4.B (Invertibility; supplement)

Let $\{X_t\}$ be a stationary solution of $\phi(B)X_t = \theta(B)\epsilon_t$ and suppose $\theta(z) \neq 0$ for $|z| \leq 1$. Then (2.20) holds a.s., with $|\pi_j|$ decaying geometrically.

Proof. [Supplement] As in Proposition 2.2.B, $1/\theta = \sum_j \eta_j z^j$ and $\pi = \phi\eta$ have geometrically decaying coefficients. By the Lemma,

$$\pi(B)X_t = \eta(B)(\phi(B)X_t) = \eta(B)(\theta(B)\epsilon_t) = (\eta\theta)(B)\epsilon_t = \epsilon_t.$$

■

Example 2.4.2 (One ACF, two MA(1) models)

The models

$$X_t = \epsilon_t + 2\epsilon_{t-1} \ \text{(not invertible)}, \qquad X_t = \epsilon_t + \tfrac{1}{2}\epsilon_{t-1} \ \text{(invertible)} \tag{2.21}$$

both have $\rho(1) = \theta/(1+\theta^2) = 2/5$ and $\rho(h) = 0$ for $|h| \geq 2$. Generally θ and $1/\theta$ give the same ACF; the ACVFs agree with noise variance σ^2 for $\theta = 2$ and $4\sigma^2$ for $\theta = 1/2$.

The invertible model is preferred because (2.20) gives the predictor

$$\hat{X}_{n+1} = -\pi_1 X_n - \pi_2 X_{n-1} - \cdots - \pi_n X_1.$$

Orders p, q are chosen from data: too many parameters violate parsimony, too few cause *model bias*; AIC and BIC are used (Chapter 4). For an AR(p) with $n > p$, by causality

$$E(X_{n+1} \mid X_1, \ldots, X_n) = E(\phi_1 X_n + \cdots + \phi_p X_{n-p+1} + \epsilon_{n+1} \mid X_1, \ldots, X_n) = \phi_1 X_n + \cdots + \phi_p X_{n-p+1},$$

so a wrong p gives a wrong predictor.

Example 2.4.3 (ACVF of a causal ARMA(1,1))

Let $X_t = \phi X_{t-1} + \epsilon_t + \theta \epsilon_{t-1}$, $|\phi| < 1$. For $h \geq 2$, X_{t-h} is uncorrelated with $\epsilon_t, \epsilon_{t-1}$, so

$$\gamma(h) = \mathrm{Cov}(\phi X_{t-1} + \epsilon_t + \theta \epsilon_{t-1}, X_{t-h}) = \phi \gamma(h-1), \quad \text{so} \quad \gamma(h) = \phi^{h-1} \gamma(1), \ h \geq 1.$$

With $\mathrm{Cov}(\epsilon_{t-1}, X_{t-1}) = \mathrm{Cov}(\epsilon_t, X_t) = \sigma^2$, $\mathrm{Cov}(\epsilon_{t-1}, X_t) = (\phi + \theta)\sigma^2$:

$$\begin{aligned} \gamma(1) &= \phi \gamma(0) + \theta \sigma^2, \\ \gamma(0) &= \phi^2 \gamma(0) + \sigma^2 + \theta^2 \sigma^2 + 2\phi\theta\sigma^2. \end{aligned}$$

Hence

$$\gamma(0) = \frac{\sigma^2(1 + 2\phi\theta + \theta^2)}{1 - \phi^2}, \qquad \gamma(1) = \frac{\sigma^2(\phi + \theta)(1 + \phi\theta)}{1 - \phi^2}.$$

2.5 The partial autocorrelation function

Motivation. The PACF removes the effect of intermediate observations and cuts off after lag p for an AR(p).

Let $\{X_t\}$ be stationary with mean 0,

$$F = \Big\{ \sum_{i=2}^{k} a_i X_i : a_i \in \mathbb{R} \Big\},$$

and $P_F Y$ the *projection* of Y onto F: $E[(Y - P_F Y) X_i] = 0$, $i = 2, \ldots, k$ (the best linear predictor; $F = \{0\}$ for $k = 1$).

Definition 2.5.1 (Partial autocorrelation function)

For $k \geq 1$,

$$\phi_{kk} = \mathrm{Corr}(X_{k+1} - P_F X_{k+1},\ X_1 - P_F X_1)$$

is the *partial autocorrelation function* (PACF); for fixed k, ϕ_{kk} is the partial autocorrelation coefficient at lag k.

2.5.1 Computing the PACF

ϕ_{hh} is the last component of the solution of

$$\begin{pmatrix} \rho(0) & \rho(1) & \cdots & \rho(h-1) \\ \rho(1) & \rho(0) & \cdots & \rho(h-2) \\ \vdots & \vdots & \ddots & \vdots \\ \rho(h-1) & \rho(h-2) & \cdots & \rho(0) \end{pmatrix} \begin{pmatrix} \phi_{h1} \\ \phi_{h2} \\ \vdots \\ \phi_{hh} \end{pmatrix} = \begin{pmatrix} \rho(1) \\ \rho(2) \\ \vdots \\ \rho(h) \end{pmatrix}. \tag{2.22}$$

By Cramer's rule, with $R_h = [\rho(i-j)]_{i,j=1}^h$ and R_h^* its last column replaced by $(\rho(1),\dots,\rho(h))^T$,

$$\phi_{hh} = \frac{\det R_h^*}{\det R_h}.$$

Proposition 2.5.A (PACF via the prediction equations; supplement)

Let $\{X_t\}$ be stationary, mean 0, with $\Gamma_h = [\gamma(i-j)]_{i,j=1}^h$ nonsingular. Then the best linear predictor of X_{h+1} from $X_1,\dots,X_h$ is $\sum_{j=1}^h \phi_{hj} X_{h+1-j}$ with coefficients solving (2.22), and its coefficient ϕ_{hh} on X_1 equals the PACF of Definition 2.5.1.

Proof. [Supplement]

Step 1. $\hat{X} = \sum_j c_j X_{h+1-j}$ is the projection iff $E[(X_{h+1} - \hat{X})X_{h+1-i}] = 0$, $i = 1,\dots,h$, i.e. $\sum_j c_j \gamma(i-j) = \gamma(i)$; divide by $\gamma(0)$ to get (2.22), uniquely solvable.

Step 2. Let $F = \operatorname{span}(X_2,\dots,X_h)$ and $U = X_1 - P_F X_1 \perp F$ ($\perp$: uncorrelated). Since $\operatorname{span}(X_1,\dots,X_h) = F \oplus \operatorname{span}(U)$, the projection of X_{h+1} is

$$P_F X_{h+1} + \frac{E(X_{h+1}U)}{EU^2}\,U = P_F X_{h+1} + \frac{E(X_{h+1}U)}{EU^2}\,(X_1 - P_F X_1),$$

and as $P_F X_{h+1}, P_F X_1 \in F$, the coefficient of X_1 is $\phi_{hh} = E(X_{h+1}U)/EU^2$.

Step 3. With $V = X_{h+1} - P_F X_{h+1}$, $E(X_{h+1}U) = E(VU)$. As γ is even, $(X_2,\dots,X_{h+1})$ and $(X_h,\dots,X_1)$ have the same covariance matrix, so $EV^2 = EU^2$ and $\phi_{hh} = E(VU)/\sqrt{EV^2 EU^2} = \operatorname{Corr}(V,U)$. ■

Example 2.5.2 (PACF of an AR(1))

For a stationary AR(1), $\rho(h) = \phi^h$ and $\phi_{11} = \rho(1) = \phi$. At $h = 2$:

$$\begin{pmatrix} 1 & \rho(1) \\ \rho(1) & 1 \end{pmatrix}\begin{pmatrix}\phi_{21}\\ \phi_{22}\end{pmatrix} = \begin{pmatrix}\rho(1)\\ \rho(2)\end{pmatrix} \implies \begin{pmatrix}\phi_{21}\\ \phi_{22}\end{pmatrix} = \frac{1}{1-\phi^2}\begin{pmatrix} 1 & -\phi \\ -\phi & 1\end{pmatrix}\begin{pmatrix}\phi\\ \phi^2\end{pmatrix} = \begin{pmatrix}\phi\\ 0\end{pmatrix},$$

so $\phi_{22} = 0$; likewise $\phi_{hh} = 0$ for all $h \geq 2$.

Proposition 2.5.B (PACF of a causal AR(p); supplement)

If $\{X_t\}$ is a causal AR(p) with $\phi_p \neq 0$ and $\sigma^2 > 0$, then $\phi_{pp} = \phi_p$ and $\phi_{hh} = 0$ for all $h > p$.

Proof. [Supplement]

Step 1. Γ_h is nonsingular: for $a \neq 0$ let i be the largest index with $a_i \neq 0$. By causality $X_i = \epsilon_i + \sum_{k\geq 1} a_k' \epsilon_{i-k}$ and $X_1,\dots,X_{i-1}$ depend on ϵ_u, $u < i$, so $Y = \sum_{j\leq i} a_j X_j = a_i\epsilon_i + W$ with $W \perp \epsilon_i$ and $a^T\Gamma_h a = \operatorname{Var}(Y) \geq a_i^2\sigma^2 > 0$.

Step 2. For $h \geq p$, $\hat{X} = \phi_1 X_h + \cdots + \phi_p X_{h+1-p}$ has error $\epsilon_{h+1} \perp X_1,\dots,X_h$ (causality), so it is the projection, with coefficients $(\phi_1,\dots,\phi_p,0,\dots,0)$.

Step 3. By Proposition 2.5.A, ϕ_{hh} is the coefficient on X_1: ϕ_p if $h = p$, 0 if $h > p$. ■

So the AR(p) PACF cuts off after lag p and the MA(q) ACF after lag q, the other functions decaying geometrically (Figure 1); this guides the choice of p, q (Chapters 3–4).

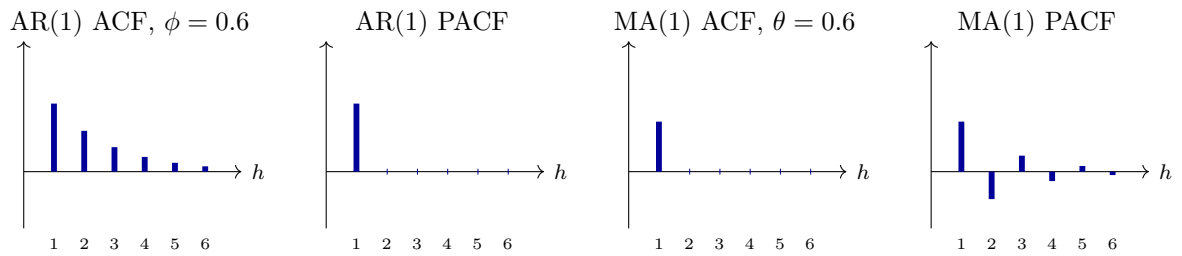


Figure 1: ACF and PACF of an AR(1) and an MA(1): the AR(1) PACF and MA(1) ACF cut off after lag 1.

2.6 Selected exercises

Motivation. Exercises on sums of MA processes, models sharing an ACF, and invertibility of ARMA(1,1).

Exercise 2.8 (Sum of independent MA(1) and MA(2))

Let $X_t = \epsilon_t + a\epsilon_{t-1}$ with $\{\epsilon_t\} \sim IID(0, \sigma_1^2)$ and $Y_t = \eta_t + b_1\eta_{t-1} + b_2\eta_{t-2}$ with $\{\eta_t\} \sim IID(0, \sigma_2^2)$, the two noise sequences being independent, and $\sigma_1^2 + \sigma_2^2 > 0$. Show that $Z_t = X_t + Y_t$ can be regarded as an MA(2) model: there are a white noise $\{u_t\} \sim WN(0, v)$ with $v > 0$ and constants c_1, c_2 such that

$$Z_t = u_t + c_1 u_{t-1} + c_2 u_{t-2} \qquad \text{for all } t.$$

Solution. Work in L^2 with inner product $E(UV)$; $\perp$ means uncorrelated. Let $H_t = \overline{\text{span}}\{Z_s : s \le t\}$ and P_t the projection onto H_t.

Step 1. By (2.15) and Exercise 1.7, Z is stationary, mean 0, $\gamma_Z = \gamma_X + \gamma_Y$, so $\gamma_Z(h) = 0$ for $|h| \ge 3$ and $\gamma_Z(0) = (1 + a^2)\sigma_1^2 + (1 + b_1^2 + b_2^2)\sigma_2^2 > 0$. Hence $Z_t \perp H_{t-3}$.

Step 2. (Innovations.) $u_t = Z_t - P_{t-1}Z_t \in H_t$, $u_t \perp H_{t-1}$; for $s < t$, $u_s \in H_{t-1}$, so $E(u_s u_t) = 0$.

Step 3. (Shift invariance.) Let α_m be the minimum-norm solution of $\Gamma_m \alpha_m = (\gamma_Z(1), \dots, \gamma_Z(m))^T$. As in Step 1 of Proposition 2.5.A, $\sum_i \alpha_{m,i} Z_{t-i}$ is the projection onto $\text{span}(Z_{t-1}, \dots, Z_{t-m})$, with t-free coefficients; these converge in L^2 to $P_{t-1}Z_t$. By continuity, $E(u_t u_s)$ and $E(u_t Z_s)$ depend only on $t - s$; so $v = Eu_t^2$ is constant and $\{u_t\} \sim WN(0, v)$.

Step 4. ($v > 0$.) If $v = 0$, $Z_s \in H_{s-1}$ for all s, so $H_t = H_{t-3}$; then $Z_t \perp H_t \ni Z_t$, i.e. $EZ_t^2 = 0$, a contradiction.

Step 5. $Z_s = u_s + P_{s-1}Z_s$ gives $H_s = H_{s-1} \oplus \text{span}(u_s)$; for $s = t-1, t-2$,

$$H_{t-1} = H_{t-3} \oplus \text{span}(u_{t-2}) \oplus \text{span}(u_{t-1}).$$

Step 6. Projecting Z_t,

$$Z_t = u_t + P_{t-1}Z_t = u_t + \frac{E(Z_t u_{t-1})}{v}\, u_{t-1} + \frac{E(Z_t u_{t-2})}{v}\, u_{t-2} + P_{H_{t-3}} Z_t,$$

the last term 0 by Step 1 and $c_j = E(Z_t u_{t-j})/v$ free of t by Step 3. Since $u_{t-2} - Z_{t-2} \in H_{t-3} \perp Z_t$, $c_2 = \gamma_Z(2)/v = b_2\sigma_2^2/v$.

Here $\{u_t\}$ is white noise, not necessarily i.i.d. ■

Exercise 2.9 (Two filters with the same ACF)

Let $\{Y_t\}$ be stationary with mean 0 and ACVF γ_Y, not a.s. zero, and let

$$X_t = Y_t - 0.4Y_{t-1}, \qquad W_t = Y_t - 2.5Y_{t-1}.$$

(a) Express the ACVFs of $\{X_t\}$ and $\{W_t\}$ through γ_Y.
(b) Show that $\{X_t\}$ and $\{W_t\}$ have the same ACF.

Solution.

(a) For $V_t = Y_t - cY_{t-1}$,

$$\begin{aligned}\mathrm{Cov}(V_t, V_{t+h}) &= \gamma_Y(h) - c\,\gamma_Y(h-1) - c\,\gamma_Y(h+1) + c^2\gamma_Y(h) \\ &= (1+c^2)\gamma_Y(h) - c\{\gamma_Y(h-1) + \gamma_Y(h+1)\},\end{aligned}$$

so with $c = 0.4$ and $c = 2.5$,

$$\begin{aligned}\gamma_X(h) &= 1.16\,\gamma_Y(h) - 0.4\{\gamma_Y(h-1) + \gamma_Y(h+1)\}, \\ \gamma_W(h) &= 7.25\,\gamma_Y(h) - 2.5\{\gamma_Y(h-1) + \gamma_Y(h+1)\}.\end{aligned}$$

(b)

Step 1. $7.25 = 6.25 \times 1.16$ and $2.5 = 6.25 \times 0.4$, so $\gamma_W = 6.25\,\gamma_X$ (generally $1 + c^{-2} = c^{-2}(1+c^2)$, $c^{-1} = c^{-2}c$).

Step 2. $\gamma_X(0) > 0$: otherwise $Y_t = 0.4Y_{t-1} = 0.4^n Y_{t-n}$ a.s., so $\gamma_Y(0) = 0.16^n\gamma_Y(0)$ for all n and $Y_t = 0$ a.s.

Step 3. Hence $\rho_W(h) = 6.25\gamma_X(h)/(6.25\gamma_X(0)) = \rho_X(h)$; this generalizes Example 2.4.2. ■

Exercise 2.10 (Invertibility of ARMA(1,1))

Let $\{X_t\}$ be the stationary solution of $X_t - \phi X_{t-1} = \epsilon_t + \theta\epsilon_{t-1}$, $\{\epsilon_t\} \sim IID(0, \sigma^2)$, $\sigma^2 > 0$, $|\phi| < 1$. Find the condition for $\{X_t\}$ to be invertible, i.e. for $\epsilon_t = \sum_{j\geq 0}\pi_j X_{t-j}$ a.s. for some $\sum_j |\pi_j| < \infty$.

Solution. The condition is $|\theta| < 1$.

(a) Sufficiency.

Step 1. If $|\theta| < 1$, the root $-1/\theta$ of $\theta(z)$ lies outside the unit disc; apply Proposition 2.4.B.

Step 2. Explicitly, for $|z| < 1/|\theta|$,

$$\frac{\phi(z)}{\theta(z)} = (1-\phi z)\sum_{k\geq 0}(-\theta)^k z^k = 1 + \sum_{j\geq 1}\{(-\theta)^j - \phi(-\theta)^{j-1}\}z^j,$$

so $\pi_0 = 1$, $\pi_j = -(\phi+\theta)(-\theta)^{j-1}$, absolutely summable.

(b) Necessity. Suppose $\epsilon_t = \pi(B)X_t$ with $\sum|\pi_j| < \infty$ and $|\theta| \geq 1$.

Step 1. By Proposition 2.4.A and Lemma 2.2.A, $X_t = a(B)\epsilon_t$ with $a = \theta/\phi$ on $|z| \leq 1$, and $\epsilon_t = c(B)\epsilon_t$ with $c = \pi a$, $\sum|c_k| < \infty$.

Step 2. Covariance with ϵ_{t-k} gives $\sigma^2\mathbb{1}\{k=0\} = c_k\sigma^2$, so $c(z) \equiv 1$ and

$$\pi(z)\,\theta(z) = \phi(z), \qquad |z| \leq 1.$$

Step 3. At $z_0 = -1/\theta$ ($|z_0| \leq 1$), $\phi(z_0) = 1 + \phi/\theta = 0$, so $|\phi| = |\theta| \geq 1$, a contradiction.

If $\phi = -\theta$ (common factor), $X_t = \epsilon_t$, consistent with the criterion. ■